A2 Q4

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4. a) Theoretical Covar:

$Cov(X,Y) = E[XY] - E[X]E[Y]$

$= E[(X-E[X])(Y-E[Y])$

Let $\hat{\beta}_0, \hat{\beta}_1$ be X, Y

We know

$\hat{\beta}_1 = \sum_{i=1}^n (x_i-\bar{x})(Y_i-\bar{Y}) / \sum_{i=1}^n (x_i-\bar{x})^2 = \sum_{i=1}^n c_i Y_i$

$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}$

$E(\hat{\beta}_1) = \sum \left(\frac{x_i-\bar{x}}{\sum_{i=1}^n (x_i-\bar{x})^2} \right) (\beta_0 + \beta_1 x_i) = \beta_1$

Let $k = \sum_{i=1}^n (x_i-\bar{x})^2$

Another way to express $Cov(\hat{\beta}_0, \hat{\beta}_1)$ is ... $= Cov(\bar{Y} - \hat{\beta}_1 \bar{x}, \hat{\beta}_1)$

By linearity of covar.: $= Cov(\bar{Y}, \hat{\beta}_1) - Cov(\hat{\beta}_1 \bar{x}, \hat{\beta}_1)$

Thus, we must derive the two.

1. $Cov(\bar{Y}, \hat{\beta}_1)$

Let $\bar{Y} = \sum_{i=1}^n \frac{1}{n} Y_i$ by independence

We know $Var(Y_i) = \sigma^2$

And we simplify $\hat{\beta}_1 = \sum_{i=1}^n (x_i-\bar{x})(Y_i-\bar{Y})/k = \sum_{i=1}^n (x_i-\bar{x})Y_i/k$ since $\sum_{i=1}^n (x_i-\bar{x})\bar{Y} = \bar{Y}\sum_{i=1}^n (x_i-\bar{x}) = 0$ ($\sum_{i=1}^n (x_i-\bar{x}) = 0$)

Let $c_i = (x_i-\bar{x})/k$

By properties of covariance, $Cov(\sum a_i Y_i, \sum b_j Y_j) = \sigma^2 \sum a_i b_i$ where $a_i = \frac{1}{n}$ and $b_i = c_i$ in this context

Thus, $Cov(\bar{Y}, \hat{\beta}_1) = \sigma^2 \sum_{i=1}^n \frac{1}{n} c_i$

$= \sigma^2 \sum_{i=1}^n \frac{1}{n} \left(\frac{x_i-\bar{x}}{k} \right)$

$= \frac{\sigma^2}{nk} \sum_{i=1}^n x_i - \bar{x}$

$= 0$

2. $Cov(\hat{\beta}_1 \bar{x}, \beta_1)$

$= \bar{x}\, Cov(\hat{\beta}_1, \hat{\beta}_1)$

$= \bar{x}\, Var(\hat{\beta}_1)$ (by property of covar.)

$= \bar{x} \left[\frac{\sigma^2}{k} \right]$

Finally, $Cov(\hat{\beta}_0, \hat{\beta}_1) = Cov(\bar{Y}, \hat{\beta}_1) - Cov(\hat{\beta}_1 \bar{x}, \beta_1)$

$= 0 - \bar{x} \left[\frac{\sigma^2}{k} \right]$

$= \frac{-\sigma^2 \bar{x}}{\sum_{i=1}^n (x_i-\bar{x})^2}$ ■

b) Let $s = \sigma^2 \left[\frac{1}{n} + \frac{(x-\bar{x})^2}{k} \right]$, using earlier k

$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 x \sim N(\beta_0 + \beta_1 x, s)$

Mean

$E[\beta_0] = \beta_0$

$E[\hat{\beta}_1] = \beta_1$

$\therefore E[\hat{Y}] = E[\hat{\beta}_0] + E[\hat{\beta}_1] x$ (x: const.)

$= \beta_0 + \beta_1 x$

Variance

We know $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}$

and we assume $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 x$ holds

thus, $\hat{Y} = \bar{Y} - \hat{\beta}_1 \bar{x} + \hat{\beta}_1 x$

$= \bar{Y} + \hat{\beta}(x - \bar{x})$

Now we calc $Var(\hat{Y}) = Var(\bar{Y} + \hat{\beta}_1(x - \bar{x}))$

by properties of variance:

$$Var(\hat{Y}) = Var(\bar{Y}) + Var(\hat{\beta}_1(x-\bar{x})) + 2\,Cov(\bar{Y}, \hat{\beta}_1(x-\bar{x}))$$

Since $(x - \bar{x})$ is a constant:

$$= Var(\bar{Y}) + (x-\bar{x})^2 Var(\beta_1) + 2(x-\bar{x})Cov(\bar{Y}, \hat{\beta}_1)$$

From a), we know $Cov(\bar{Y}, \beta_1) = 0$

$$= Var(\bar{Y}) + (x-\bar{x})\,Var(\beta_1)$$

We know $\bar{Y} = \sum_{i=1}^{n} Y_i/n$, and since $Var(Y_i) = \sigma^2$, $Var(\bar{Y}) = Var\left(\sum_{i=1}^{n} \frac{Y_i}{n}\right) = \frac{\sigma^2}{n}$

$$= \frac{\sigma^2}{n} + \frac{\sigma^2}{k}(x-\bar{x})$$

$$= \sigma^2\left[\frac{1}{n} + \frac{(x-\bar{x})}{\sum_{i=1}^{n}(x-\bar{x})^2}\right]$$

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